

1. Find the derivatives (if they exist) of

$$f(x) = \begin{cases} x^2 e^{-x^2} & \text{if } |x| \leq 1 \\ \frac{1}{e} & \text{if } |x| > 1 \end{cases}, \quad g(x) = |x^2 - 4|$$

2. Let $g(x) = |x^3|$ for $x \in \mathbb{R}$. Determine whether $g'(0)$, $g^{(2)}(0)$, $g^{(3)}(0)$ exist and if yes find them.

3. Let $f(x) = e^{2x} \sin(x)$.

(a) Find the first derivative, $f'(x)$.

(b) Compute the second derivative, $f''(x)$.

(c) Derive a general expression for $f^{(n)}(x)$, the n -th derivative of $f(x)$.

4. Let $f(x) = x^3 - 3x^2 + 4x$ on the interval $[0, 2]$.

(a) Verify that $f(x)$ satisfies the conditions of the Mean Value Theorem on $[0, 2]$.

(b) Find the value of c in $(0, 2)$ such that $f'(c) = \frac{f(2) - f(0)}{2 - 0}$.

5. Given the equation $x^2 + y^2 = 25$,

(a) Use implicit differentiation to find $\frac{dy}{dx}$.

(b) Determine the slope of the tangent line at the point $(3, 4)$.

6. Let $f(x) = \ln(x)$.

(a) Find the equation of the tangent line to $f(x)$ at $x = 2$.

(b) Find the equation of the normal line to $f(x)$ at $x = 2$.

7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^2, & x \in \mathbb{Q} \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

(a) Show that f is differentiable at $c = 0$ and find $f'(0)$.

(b) Prove that if $c \neq 0$ then f is not differentiable at c .

8. Let f be a differentiable function at 0 such that $f(0) = 0$. For all $k \in \mathbb{N}$, find

$$\lim_{x \rightarrow 0} \frac{1}{x} \left(f(x) + f\left(\frac{x}{2}\right) + f\left(\frac{x}{3}\right) + \dots + f\left(\frac{x}{k}\right) \right).$$

9. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at $c \in \mathbb{R}$, explain why

$$f'(c) = \lim_{n \rightarrow \infty} [n(f(c + 1/n) - f(c))]$$

Give an example of a function f and a number c such that

$$\lim_{n \rightarrow \infty} [n(f(c + 1/n) - f(c))]$$

exists but $f'(c)$ does not exist.

10. Use the Mean Value Theorem to show that

$$|\cos(x) - \cos(y)| \leq |x - y|.$$

In general, suppose that $f : [a, b] \rightarrow \mathbb{R}$ is such that f' exists on $[a, b]$ and f' is continuous on $[a, b]$. Prove that f is Lipschitz on $[a, b]$.

11. Give an example of a uniformly continuous function on $[0, 1]$ that is differentiable on $(0, 1)$ but whose derivative is not bounded on $(0, 1)$. Justify your answer.
12. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) . We proved that if $f'(x) = 0$ for all $x \in (a, b)$ then f is constant on $[a, b]$. Give an example of a function $f : A \rightarrow \mathbb{R}$ such that $f'(x) = 0$ for all $x \in A$ but f is not constant on A .
13. Let $f(x) = x^3 - 6x^2 + 9x + 1$.
- (a) Find the critical points of f .
 - (b) Determine whether each critical point corresponds to a local maximum, a local minimum, or neither.
14. Let $f(x) = x^4 - 4x^2 + 3$ on the interval $[-2, 2]$.
- (a) Find the critical points of f .
 - (b) Evaluate $f(x)$ at the critical points and the endpoints of the interval.
 - (c) Determine the global maximum and minimum values of f on $[-2, 2]$.
15. Let $f(x) = e^{-x^2}$.
- (a) Find the critical points of f .
 - (b) Use the second derivative test to classify the nature of each critical point.
16. A rectangular field is to be fenced on three sides (two lengths and one width), leaving one side open. The total length of the fence available is 60 meters.
- (a) Express the area of the field as a function of one variable.
 - (b) Determine the dimensions of the field that maximize the area.
 - (c) Find the maximum area.
17. Let $f(x) = \sin(x) + \cos(x)$ on the interval $[0, 2\pi]$.
- (a) Find the critical points of f .
 - (b) Determine the maximum and minimum values of f on $[0, 2\pi]$.
18. Establish the inequalities for $x > 0, y > 0, p > 0, q > 0$ such that $\frac{1}{p} + \frac{1}{q} = 1$
- (a) $xy \leq \frac{x^p}{p} + \frac{y^q}{q}$, (b) $(x + y) \ln\left(\frac{x+y}{2}\right) \leq x \ln x + y \ln y$.
19. Prove that , for $x_k \geq 0$.

$$(a) \sqrt[n]{\prod_{k=1}^n x_k} \leq \frac{1}{n} \sum_{k=1}^n x_k, \quad (b) \frac{\exp b - \exp a}{b - a} \leq \frac{\exp a + \exp b}{2}.$$